Econ 201B: Lecture 4

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1 Domination Theorem Rephrased

Theorem 1. *In a finite game, a strategy is dominated iff there is no belief to which it is a best response*

In other words: if $\sigma_i \in \Delta(S_i)$ is not dominated, then there are beliefs μ s.t. $\sigma_i \in \beta_i(\mu)$.

1.1 IEDS (Iterated Elimination of Dominated Strategies)

Let $(N, (S_i), (\mu_i))$ be a finite game $N = [n]$. Define n sequences of sets of mixed strategies. Let $D_i^0 = \Delta(S_i)$ be all mixed strategies. Given $D_i^{k-1}, ..., D_n^{k-1}$ let

$$D_i^k = \left\{\sigma_i : \nexists \bar{\sigma}_i : u_i(\sigma_i, \sigma_{-i}) < u_i(\bar{\sigma}_i, \sigma_{-i}) \, \forall \sigma_{-i} \in \times_{j \neq i} D_j^{k-1}\right\}$$

start with strategy check with dominated strategy eliminated given other players' mixed strategies in $\times_{j \neq i} D_j^{k-1}$. Where D_i^k are strategies that have not been eliminated from step k.

Let $D_i = \cap_{k=0}^{\infty} D_i^k$, set of strategies that never get eliminated. A game is called dominant solvable if D is a singleton. With strict dominance and finite # of players, order does not matter but this is not true with weak dominance!

1.2 Rationalizability

Based on keeping the best responses to some beliefs instead of relying on dominance.

$$R_i = \Delta(S_i)$$

Given $R_1^{k-1}, ..., R_n^{k-1}$, let

$$Z_i^k = \left\{s_i \in S_i : \sigma_i(s_i) > 0 \text{ for some } \sigma_i \in R_i^{k-1}\right\}$$
$$R_i^k = \left\{\sigma_i \in \Delta(S_i) : \exists \mu \in \Delta\left(\times_{i \neq j} Z_j^k\right) \text{ s.t. } \sigma_i \in \beta_i(\mu)\right\}$$

Support of beliefs for other player has to be positive, i.e., strategies must have survived so far $\times_{i \neq j} Z_j^k$. We also allow for beliefs to certain some correlation.

The rationalizable strategies are the elements of $R = \times_{i=1}^n R_i$. In a finite game, R is always non-empty and contains at least 1 pure strategy profile.

Proposition 2. *$\sigma_i \in \Delta(S_i)$ is a rationalizable if there are sets $Z_1, ..., Z_n$ where $Z_j \subseteq S_j$ s.t.*

1. $\sigma_i \in \beta_i(\mu)$ for some $\mu_i \in \Delta(\times_{i \neq j} Z_n)$
2. For every $s_j \in Z_j$ there is $\mu_j \in \Delta(\times x_{i \neq j} Z_j)$ s.t. $s_j \in \beta_j(\mu_j)$.

As a consequence of domination theorem

$$R = D$$

2 Nash Equilibrium

There is a difference between rationalizability and Nash Equilibrium.

- R: Rationality and rationalizable "beliefs" where beliefs can be incorrect
- NE: Rationality and "correct" beliefs where agents have perfect foresight. Common knowledge, which justifies rationality, has nothing to do with equilibrium

Definition 3. A strategy profile σ is a NE if no player has a profitable deviation. Equivalently, σ is a NE iff

$$\begin{aligned} \sigma &\in \arg\max_{\sigma_i} u_i\left(\sigma_i', \sigma_{-i}\right) \\ \text{s.t.} \sigma_i' &\in \Delta\left(S_i\right) \end{aligned}$$

A σ_i solve the linear program iff every pure strategy in its support solves the maximization problem. Nash Equilibrium strategies have the following properties: all pure strategies in the support of σ_i yield the same utility; in any finite game, there exists a finite and odd number of equilibrium – this statement is generically true; and if σ is a NE profile, the σ_i cannot be dominated – in fact, σ is rationalizable.